Answers are at the end of the chapter.

10.2.8 DNS rebinding attacks

A more sophisticated SSRF attack, which can defeat validation of redirects, is a *DNS rebinding attack*, in which an attacker sets up a website and configures the DNS server for the domain to a server under their control. When the validation code looks up the IP address, the DNS server returns a genuine external IP address with a very short time-to-live value to prevent the result being cached. After validation has succeeded, Jsoup will perform another DNS lookup to actually connect to the website. For this second lookup, the attacker's DNS server returns an internal IP address, and so Jsoup attempts to connect to the given internal service.

> DEFINITION A ***DNS rebinding*** attack occurs when an attacker sets up a fake website that they control the DNS for. After initially returning a correct IP address to bypass any validation steps, the attacker quickly switches the DNS settings to return the IP address of an internal service when the actual HTTP call is made.